MIL-D-55456B(EL)
26 June 1969
SUPERSEDES
MIL-D-55456A(EL)
2 May 1967

MILITARY SPECIFICATION

DISTRIBUTION BOX, J-1077( )/U
DISTRIBUTION BOX, J-2317( )/U

1. SCOPE

1.1 Scope.- This specification covers two types of distribution boxes designated as Distribution Box J-1077( )/U and Distribution Box J-2317( )/U.

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-S-760	Strapping, Non-metallic (and Connectors).
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-B-43291	Boxes, Fiberboard, Corrugated, Double-Wall, Weather Resistant.
MIL-C-55074	Connector, Plug, Electrical U-185( )/G, Connector, Receptacle, Electrical U-186( )/G, Connector, Receptacle, Electrical U-187( )/G, and Contact Assembly, Electrical MX-3227( )/G.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads 40 inches by 48 inches 4-way Partial and 4-way Pallets.
MIL-STD-169	Extreme Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-173820	Connector, Receptacle, Electrical U-187( )/G.
SC-DL-288038	Distribution Box J-1077( )/U
SC-GL-323007	Electrical Fixture for U-185( )/G, U-186( )/G, U-187( )/G Connectors and Contact Assembly MX-3227( )/G.
SC-DL-555825	Distribution Box J-2317( )/U.

(Copies of specifications, standards, and drawings, required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- The equipments which this specification covers are Distribution Boxes J-1077( )/U and J-2317( )/U which are used with area type communication system utilizing 26 pair cable assemblies.

3.2 Construction.- The equipment shall be constructed in accordance with the requirements of this specification and the following Drawing and Data Lists:

SC-DL-173820	Connector, Receptacle, Electrical U-187( )/G.
SC-DL-288038	Distribution Box J-1077( )/U.
SC-DL-555825	Distribution Box J-2317( )/U.

3.3 Cleaning.- After assembly, the equipment shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

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3.4 Finish, protective.- Equipment shall be given protective finish in accordance with MIL-F-14072 and the equipment drawings. (See 4.3).

3.5 Marking.- Marking shall conform to MIL-M-13231. (See 4.3).

3.6 Connectors.- Connector, Receptacle, Electrical U-187( )/G shall meet the requirements of MIL-C-55074. (See 4.11).

3.7 Electrical requirements.-

3.7.1 Continuity.- Each circuit shall be electrically continuous and the resistance shall not exceed 1 ohm. (See 4.7.1).

3.7.2 Dielectric strength.- The equipment shall be considered a reject if the automatic tester fails to proceed after a period of 1 minute when tested in accordance with 4.7.2.

3.7.3 Insulation resistance.- When tested in accordance with 4.7.3, the insulation resistance shall be not less than 1,000 megohms except during the Moisture resistance test 4.9.4 and the High humidity and high temperature test 4.9.5 when it shall be not less than 10 megohms.

3.8 Mechanical requirements.-

3.8.1 Durability.- After being subjected to the test specified in 4.8.1, each binding post shall operate freely and a tinned conductor from WM-130( )/U cable, inserted in each binding post terminal, shall withstand a 10 pound pull.

3.9 Service conditions.- The equipment shall meet the following service conditions.

3.9.1 Shock drop (J-1077( )/U only).- After being subjected to the test specified in 4.9.1, the equipment shall meet the electrical requirements of 3.7 and any physical damage shall be minor.

3.9.2 Vibration.- After being subjected to the test specified in 4.9.2, the equipment shall meet the electrical requirements of 3.7 and any physical damage shall be minor.

3.9.3 Rain.- After being subjected to the test specified in 4.9.3, the equipment shall meet the electrical requirements of 3.7 and there shall be no evidence of water leakage in the distribution box through the gaskets and/or in the housings of the connectors.

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3.9.4 Moisture resistance.- After being subjected to the test specified in 4.9.4, the equipment shall show no evidence of corrosion and shall meet the requirements of 3.7.3 Insulation resistance.

3.9.5 High humidity and high temperature.- During the test specified in 4.9.5, the equipment shall meet the requirements of 3.7.3 Insulation resistance and there shall be no evidence of corrosion.

3.9.6 Temperature cycling.- Following the test specified in 4.9.6, the equipment shall meet the electrical requirements of 3.7 and there shall be no evidence of deterioration.

3.10 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portion of the following paragraph: (See 4.10).

3.3 Cleaning

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility of inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Inspection covered by subsidiary documents. (See 4.3).

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.4).

(2) Quality conformance inspection of preparation for delivery. (See 4.12).

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4.3 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

4.4 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.4.1 Group A inspection.- Each unit (100%) of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQLs indicated in table I. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table I - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Electrical				
Continuity	3.7.1	4.7.1	.65% inspection by automatic means	
Dielectric strength	3.7.2	4.7.2		
Insulation resistance	3.7.3	4.7.3		
Visual and mechanical	3.10	4.10	1.0%	4.0%

4.4.2 Group B inspection.- Not required.

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4.4.3 Group C inspection.- This inspection shall consist of the tests specified in tables II, III and IV and shall be performed on samples that have been subjected to and met group A inspection. Sample units shall be selected at random without regard to their quality except that the samples selected at the start of production shall be selected from the first units produced.

4.4.3.1 Group C-1 inspection.- This inspection shall consist of the tests specified in table II and shall be performed as follows:

J-1077( )/U - 2 units per month.
J-2317( )/U - 1 unit per month.

Table II - Group C-1 inspection

Inspection	Req. Para.	Insp. Para.
Durability	3.8.1	4.8.1

4.4.3.2 Group C-2 inspection.- This inspection shall consist of the tests specified in table III and shall be performed on one unit every 6 months. More than one unit may be used to expedite testing.

Table III - Group C-2 inspection

Inspection	Req. Para.	Insp. Para.
Shock drop	3.9.1	4.9.1
Vibration	3.9.2	4.9.2
Rain	3.9.3	4.9.3
High humidity and high temperature	3.9.5	4.9.5
Temperature cycling	3.9.6	4.9.6

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4.4.3.3 Group C-3 inspection.- This inspection shall consist of the test specified in table IV and shall be performed on 1 unit once each year.

Table IV - Group C-3 inspection

Inspection	Req. Para.	Insp. Para.
Moisture resistance	3.9.4	4.9.4

4.4.3.4 Noncompliance.- The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and group A inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C reinspection results have shown that the corrective action was effective.

4.4.4 Reinspection of conforming group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5 Automatic testing.- The inspection for Continuity 4.7.1, Dielectric strength 4.7.2 and Insulation resistance 4.7.3 shall be performed on all equipments by automatic means using electrical fixtures specified on SC-GL-323007. Automatic test equipment other than that referenced on SC-GL-323007 shall have prior Government approval. The automatic test equipment as specified is programed to test each terminal in turn with the remaining terminals connected together to ground for dielectric strength and insulation resistance simultaneously. The automatic test equipment is also programed to test for continuity sequentially--terminals 1A thru 26A, 1B thru 26B with the read out on the tester showing 1 thru 52.

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4.6 Preconditioning.- The contractor shall be permitted to precondition the distribution box prior to performing any electrical tests except the electrical test performed during the Moisture resistance test 4.9.4 and the High humidity and high temperature test 4.9.5. The preconditioning shall consist of removing surface moisture from the equipment by wiping, drying, blowing or heating. The application of heat shall be limited to 5 minutes. For those equipments subjected to Rain 4.9.3, Moisture resistance 4.9.4 and High humidity and high temperature 4.9.5 a 24 hour waiting period in an ambient room temperature is permitted. However, if the contractor exercises this option of preconditioning, the contractor shall precondition all samples selected by Quality Assurance Representatives immediately prior to testing.

4.7 Electrical tests.-

4.7.1 Continuity.- Electrical continuity of each circuit of the distribution box shall be verified by a point-to-point test from each binding post by automatic means (See 4.5) to determine conformance with the requirements of 3.7.1.

4.7.2 Dielectric strength.- The distribution box shall be tested by automatic means. (See 4.5). The 1,000 volts supplied by the tester shall be applied instantaneously and the sequence of test shall be as fast as the tester will permit. The distribution box shall meet the requirements of 3.7.2.

4.7.3 Insulation resistance.- The distribution box shall be tested by automatic means. (See 4.5). The test will be performed with a voltage of 1,000 volts as supplied by the tester, except during the Moisture resistance test 4.9.4 and High humidity and high temperature test 4.9.5, the voltage shall be 100 volts. The distribution box shall meet the requirements of 3.7.3.

4.8 Mechanical tests.-

4.8.1 Durability.- Each binding post in the distribution box shall be subjected to 200 depressions and relaxations to determine compliance with the requirements of 3.8.1.

4.9 Service tests.-

4.9.1 Shock drop (J-1077( )/U only).- The distribution box shall be dropped a distance of 4 feet on each corner, a total of 8 drops. All drops shall be made with the cover closed. The floor or barrier receiving the impact shall be 2 inch fir backed by concrete or rigid steel frame. The distribution box shall meet the requirements of 3.9.1.

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4.9.2 Vibration.- One distribution box shall be mounted securely to a plate of the vibration table. The cover of the distribution box shall be strapped securely to the table to prevent secondary vibration. The distribution box shall be subjected to a simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion), the frequency being varied uniformly between the approximate limits of 10 and 55 cycles per second. The entire frequency range, from 10 to 55 cps and return to 10 cps shall be traversed in approximately 1 minute. The motion shall be applied for a period of 30 minutes in each of 3 mutually perpendicular directions (total of 90 minutes) to determine compliance with the requirements of 3.9.2.

4.9.3 Rain.- The distribution box, standing on end so that the cable entry slots are on the bottom, and all covers in place and secured, shall be subjected to the following:

(a) Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65°F ± 5° and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

(b) With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against 1 side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against 1 side of the equipment. Maintain this condition for 1 hour.

(c) Rotate the equipment 90 degrees horizontally to expose a second side of the equipment to the wind and rain. Maintain this condition for 1 hour.

(d) Continue as in (c) above until all 4 sides of the equipment have been exposed to the wind and rain, each side for 1 hour.

(e) Following the conditions outlined above, the equipment shall meet the requirements of 3.9.3.

4.9.4 Moisture resistance.- The distribution box, standing on end so that the cable entry slots are on the bottom, shall be subjected to 15 continuous 48-hour cycles per MIL-STD-170. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to the standard. Insulation resistance measurements shall be made during the periods shown on the standard. The equipment shall meet the requirements of 3.9.4.

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4.9.5 High humidity and high temperature.- The distribution box, standing on end so that the cable entry slots are on the bottom, shall be placed in a chamber at 86°F ± 5° and a relative humidity of 92 to 98 percent. The temperature and humidity shall be maintained continuously for 5 days. While still at these conditions, the insulation resistance shall be measured on the last day and the equipment shall meet the requirements of 3.9.5.

4.9.6 Temperature cycling.- The distribution box shall be subjected to the extreme temperature cycle shown on MIL-STD-169 and meet the requirements of 3.9.6.

4.10 Visual and mechanical inspection.- The distribution box shall be examined for the defects listed in table V. (See 3.10).

Table V - Classification of visual and mechanical defects

Classification	Defect
Major	Binding post caps missing, cracked or broken.
	Binding posts inoperative.
	Cuts or abrasions of gasket.
	Threads stripped.
	Sharp edges.
	Connector chipped or broken.
Minor	Screws missing.
	Finish not as specified.
	Marking incorrect.
	Scratches, cuts, abrasion with exposure of bare metal.
	Finish adhesion-flaking, peeling or chipping.
	Improper lead of wires.
	Taut wire.
	Insulation damaged.
	Broken strands.
	Insulation back more than 1/16 inch.

4.11 Connectors.- Connectors, Receptacle, Electrical U-187( )/G shall be tested prior to assembly on the distribution boxes to determine conformance with the requirement of 3.6.

4.12 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected to determine compliance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified. (See 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Distribution Boxes J-1077( )/U and J-2317( )/U shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying.- Distribution Boxes J-1077( )/U and J-2317( )/U shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packaging.- Distribution Boxes J-1077( )/U and J-2317( )/U. Each distribution box shall be packaged Method III in accordance with MIL-P-116 as follows: Cushion the unit with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, variety SW, class Domestic, type CF, designed to protect the unit from the shock of impact in handling and transit. Place the cushioned unit within a close-fitting box conforming to PPP-B-636, type CF, weather-resistant, style optional, special requirements. Close the box in accordance with the appendix of the box specification. In addition, seal all seams and joints with tape, not less than 2 inches wide, conforming to PPP-T-76.

5.1.2 Level C.- Distribution Boxes J-1077( )/U and J-2317( )/U shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first, receiving activity. This package may conform to the suppliers commercial practice for retail distribution when such meets the requirements of this level.

5.2 Packing.- Packing shall be level A, B, or C as specified. (See 6.2).

5.2.1 Level A.- A quantity of like stock numbered Distribution Boxes J-1077( )/U and J-2317( )/U, packaged as specified in 5.1, shall be packed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having 2 sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated to type SF, class weather-resistant, grade W5s, fiberboard conforming to PPP-F-320. The load shall be "bonded" to the pallet by strapping. When quantities per destination are less than a pallet load, packaged items shall be packed within a close-fitting style E, grade A, class 2 triple wall fiberboard box; style RSC, type 1, class 2, fiberboard box, or style RSC double-wall fiberboard box conforming to PPP-B-636, or MIL-B-43291, respectively. Closure shall be in accordance with the appendix to the box specification. Reinforcing shall be of pressure sensitive filament or non-metallic tape or non-metallic banding conforming to PPP-T-97 or PPP-S-760 respectively. Selection of the material and application shall be determined by the requirements of the appendix to the applicable box specification.

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5.2.2 Level B.- A quantity of stock numbered Distribution Boxes J-1077( )/U and J-2317( )/U, packaged as specified in 5.1, shall be packed as cited in level A except that fiberboard caps shall be class Domestic and fiberboard boxes limited to PPP-B-636 and PPP-B-640, class Domestic and class 1, respectively.

5.2.3 Level C.- A quantity of stock numbered Distribution Boxes J-1077( )/U and J-2317( )/U, packaged as specified in 5.1, shall be packed as cited in level A except that pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carriers as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The distribution boxes covered by this specification are intended for use in military communication systems to connect units of an area type communication system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Levels of preservation, packaging, packing and marking. (See section 5).

(d) Marking and shipping of samples.

(e) Place of final inspection.

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6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example, J-2317W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

CUSTODIAN

ARMY EL

Preparing Activity

ARMY EL

Project No.6110-A051

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-55436(EL) DISTRIBUTION BOX, J-3077( )/U
DISTRIBUTION BOX, J-3077( )/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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